

Surgical Cricothyrotomy

Indications for Surgical Cricothyrotomy

- Failure of 2 attempts per credentialed provider at oral intubation in patient in need of emergent airway intervention (all attempts should have utilized alternative methods – different provider, bougie, video laryngoscope, etc.)

AND

- Unable to secure airway with alternate airway device

AND AT LEAST ONE OF THE FOLLOWING:

- Inability to ventilate
- Inability to oxygenate
- Inability to clear obstructed airway in timely manner

*****OR*****

- Massive facial or oral trauma preventing BVM, intubation, or alternate airway placement as indicated by tactical situation

Contraindications

- Age < 10 years old
- Unable to find landmarks due to abnormal neck anatomy or trauma

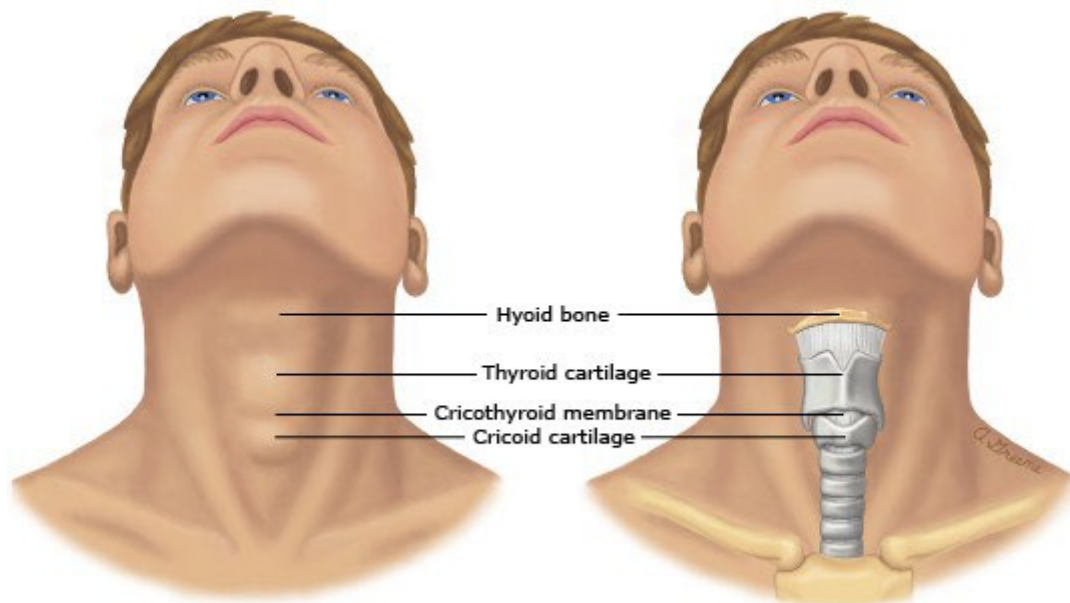
Relative Contraindications

- Able to ventilate successfully with BVM or other alternative method

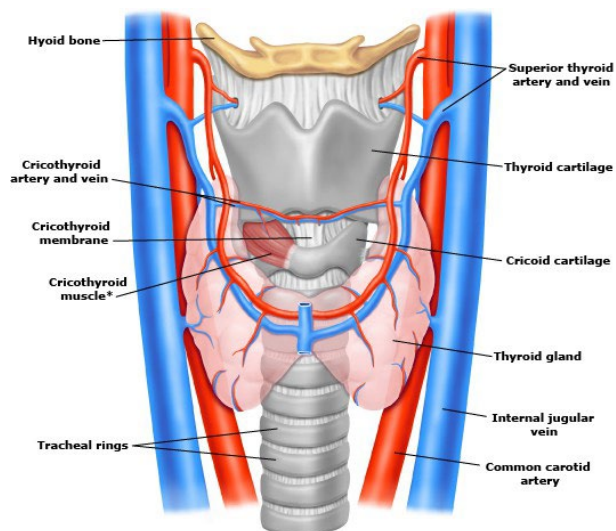
Equipment needed for surgical cricothyrotomy

- Scalpel # 11 or #15 blade
- Cricothyrotomy kit - with bougie (tube introducer) and/or tracheal hook
- ET tube size 6.0-6.5 (may be cut down to level just above pilot balloon inflation tubing and then reattach BVM connector)
- Suction
- Oxygen
- BVM
- Method to secure tube

Anatomy for surgical cricothyrotomy



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Surgical Cricothyrotomy Procedure

- 1) Prepare suction and cricothyrotomy equipment.
- 2) Place patient into neutral to sniffing position unless there is concern for trauma to c-spine.
- 3) Oxygenate the patient with 100% O₂ via NRB or BVM as appropriate.
- 4) Clean area with sterilizing prep, time permitting.
- 5) Identify the thyroid cartilage and stabilize it with your fingers until intubated.
- 6) Using a finger, identify cricothyroid membrane just below the thyroid cartilage, and above the cricoid cartilage.
- 7) Using a scalpel, make a 2-3 cm midline vertical incision through the skin over the cricothyroid membrane.
- 8) Bluntly dissect skin and tissue to expose the cricothyroid membrane.
- 9) Make a horizontal incision through the cricothyroid membrane into the trachea, being careful not to puncture through the trachea and into the esophagus.
- 10) Spread incision with finger or hemostats if needed.
- 11) Pass a bougie through the incision, feeling for tracheal rings. Pass a 6.0-6.5 ET tube over bougie and into the trachea until the cuff of the tube is just past the incision. Alternatively use the tracheal hook to secure the opening and pass the ET tube directly into the trachea.
- 12) Inflate the ET tube cuff and remove the bougie if used.
- 13) Secure tube in place and begin to ventilate with BVM and 100% O₂.
- 14) Confirm tube placement similar to orotracheal intubation, with capnography, color change device, bilateral chest rise/breath sounds, misting in the tube, and good oxygen saturation and EtCO₂.
- 15) Continuously monitor placement and functioning of tube.